Portfolio Optimization with Monte Carlo Simulations: Practical Allocation Model

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1. Summary - This project develops a quantitative portfolio optimization framework designed to construct portfolios that are both theoretically grounded and implementable, while incorporating real-world constraints and regime-dependent risk dynamics.

The model estimates expected return and risk from historical market data and improves the stability these estimates by applying Marchenko-Pastur random matrix filtering. This reduces the impact of noise in empirical correlations and leads to more stable portfolio allocations.

Using these inputs, the system solves a constrained mean-variance optimization problem, allocating capital across equities, cash, and Treasury-bills while respecting real-world constraints such as position limits and liquidity requirements.

To extend beyond static risk estimation, the model incorporates a regime-switching Monte Carlo simulation based on a Hidden Markov Model (HMM). This allows the portfolio to be evaluated under changing market conditions rather than a single, stable return distribution. This produces a full distribution of possible outcomes, including downside scenarios and probability of loss.

Overall, this model combines statistical estimation, optimization, and simulation into a single system that bridges the gap between theoretical portfolio construction and real-world investment decisions. As it integrates established portfolio theory with modern estimation and simulation techniques it includes outlines for a path of future quantitative enhancements.

2. Motivation - Traditional portfolio construction methods, particularly those based on Modern Portfolio Theory, rely on static estimates of return and risk and assume a stable covariance structure across time. In practice, these assumptions are often violated.

First, empirical correlation matrices are highly sensitive to noise, especially when the number of assets is large relative to the available data. This can lead to unstable or unintuitive portfolio allocations, as optimization procedures may overfit spurious relationships in the data.

Second, standard models typically assume a single, stationary return distribution. Financial markets, however, are inherently dynamic and exhibit regime-dependent behavior, where volatility, correlation, and return characteristics shift over time.

Third, many theoretical models produce continuous allocations that do not reflect real-world investment constraints such as position limits, liquidity needs, or the role of low-risk assets like cash and Treasury bills.

This project addresses these limitations by integrating 3 key improvements: correlation matrix denoising using random matrix theory, constrained portfolio optimization with practical allocation rules, and regime switching Monte Carlo simulations. Together, these components create a framework that is both more stable statistically and more realistic from an investment perspective.

3. Methodology

3.1 Model Inputs - This model operates on a structured set of user inputs defining the investment universe, portfolio constrains, and simulation parameters.

At the asset level, the portfolio manager specifies a set of equities along with the maximum allowable position sizes per ticker. At the portfolio level the portfolio manager defines the total capital, a historical lookback window, and a risk-aversion parameter that governs the tradeoff between return and risk. The risk aversion parameter is measured using the below equation:

$$U(w) = \mu^T w - \frac{1}{2}\lambda w^T \Sigma w$$

The segments $\mu^T w$ = expected return, $w^T \Sigma w$ = portfolio variance (risk), and λ = risk aversion parameter. Think of λ, or risk aversion, as how much return you are willing to give up reducing risk. So low risk aversion is a more aggressive investor prioritizing return, accepting higher volatility, and a more concentrated portfolio. As λ decreases risk penalty weakens, the optimizer allocates more high-returning stocks, accepts higher volatility, and may concentrate in fewer names. A higher risk aversion value is conservative investor prioritizing stability, penalizing volatility heavily, shifting toward more cash and T-bills. In the model, it represents a portfolio variance penalty increasingly, reduces equity exposure, increases cash, and diversifies more as λ increases.

In addition to defining the investment universe and core portfolio parameters, the model incorporates several inputs that govern allocation constraints and defensive positioning.

At the portfolio level, the portfolio manager specifies minimum and maximum cash allocations, ensuring that the portfolio maintains a required level of liquidity while preventing excessive idle capital. A cash yield parameter is also including this allows uninvested capital to earn a return rather than being treated as a zero-yield residual. All user inputs that represent percentages will be put in a whole number not decimal form.

The model further incorporates a Treasury-bill sleeve, which serves as a low-risk alternative to holding excess capital in cash. While cash provides liquidity, Treasury bills allow the portfolio to earn a return on defensive capital. The model can either use a user-defined Treasury bill yield or automatically retrieve a market-based rate when available. A maximum allocation constraint is applied to this sleeve to prevent over-allocation to low-risk assets while allowing the optimizer to reduce exposure to equities when risk-adjusted returns are unattractive.

In this model, the defensive portion of the portfolio is treated as a combined allocation across cash and Treasury bills rather than as separate independent decisions This allows the optimizer to

determine the most efficient mix of liquidity and low-risk yield while maintaining required capital preservation.

The model also includes flexibility in how asset allocation constraints are handled. In manual mode, the portfolio manager directly specifies maximum weights for each asset. In automatic mode, the model generates allocation caps based on asset characteristics such as volatility and correlation. Assets with lower volatility and stronger diversification properties are allowed slightly higher allocations, while more volatile or highly corelated assets receive tighter limits. This creates a balance between diversification and return-seeking behavior without requiring the portfolio manager to manually tune each constraint.

Additional inputs define the statistical and simulation environment. The lookback window determines the amount of historical data used to estimate returns and risk. A longer lookback period generally produces more stable estimates, while a shorter window allows the model to respond more quickly to recent market conditions.

The portfolio manager also defines parameters for the Monte Carlo simulation, including the number of simulation paths and the investment horizon. These inputs control the depth and time frame of the forward-looking analysis and directly impact the stability and interpretability of the simulated outcome distribution.

Finally, the model support optional portfolio targets such as a minimum expected return or a maximum allowable volatility. These constraints allow the portfolio manager to guide the optimization toward specific performance or risk objectives, ensuring that the resulting portfolio aligns with defined investment goals.

Taken together, these inputs define the feasible investment space and impose realistic constraints on the optimization process. Rather than producing purely theoretical allocations, the model translates user-defined preferences and market data into portfolios that are both statistically grounded and practically implementable.

3.2 Return and Risk Estimation – The model estimates expected return and risk directly from historical adjusted-close price data. This approach allows the optimizer to rely on observable market information rather than subjective assumptions.
Daily log returns are computed for each asset using the following expression:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

P_t represents the adjusted closing price at time t. Log returns are used because they are time-additive and provide a consistent framework for measuring returns over multiple periods.

For each asset, expected return is estimated at the average of historical daily log returns, annualized using the standard approximation of trading days per year:

$$\mu_i = \mathrm{mean}(r_i) \times 252$$

This produces an annualized expected return for each stock based on the selected lookback window.

Risk is measured using the covariance structure of asset returns. The model computes the covariance matrix from the same return series and annualizes it as shown:

$$\Sigma = cov(r) \times 252$$

The diagonal elements of the covariance matrix represent individual asset variances, while the off0diagonal elements capture the relationships between assets. These relationships are critical for measuring diversification effects within the portfolio universe.

From the covariance matrix, the model derives annualized volatility for each asset:

$$\sigma_i = \sqrt{\Sigma_{ii}}$$

Volatility serves as the primary measure of standalone asset risk, while the full covariance matrix determines how risk behaves at the portfolio level.

In addition to covariance, the model computes the correlation matrix to better understand the degree of corelating movement between assets. Correlation plays a central role in portfolio construction, as it determines how effectively risk can be reduced through diversification.

While these estimates are based on historical data, they provide a practical and widely used foundation for modeling expected return and risk. These outputs serve as the core inputs for the optimization process, which is further refined through correlation denoising in the next stage of the model. Factor-based return models provide a more structure approach to expected-return estimation and representation a natural extension beyond the estimators tested in this model.

3.3 Correlation Denoising (Marchenko-Pastur) - One the of main limitations of traditional mean-variance optimization is its sensitivity to estimation error, particularly in the correlation matrix. When the number of assets is large relative to the number of observations, empirical correlations tend to contain a significant amount of noise. This noise can lead to unstable or unrealistic portfolio allocations, as the optimizer may overfit relationships that are not truly persistent.

To address this issue, the model applies a denoising procedure based on the Marchenko-Pastur distribution from random matrix theory, this approach provides a theoretical framework for distinguishing meaningful structure from random noise in large correlation matrices.

Let *T* represent the number of time observations and *N* represent the number of assets. Defined as:

$$Q = \frac{T}{N}$$

Under the assumption of random, uncorrelated data, the eigenvalues - a scalar were multiplying a square matrix by a non-zero vector scales the vector without rotating it defined as $Av = \lambda v$) - of the correlation matrix are expected to fall within a specific range. The upper bound of this range is given by:

$$\lambda_+ = \left(1 + \frac{1}{\sqrt{Q}}\right)^2$$

Eigenvalues that fall within this range are considered to be dominated by noise, while eigenvalues above this threshold are more likely to represent meaningful market structure.

The model implements the following steps listed:

1. Compute the empirical correlation matrix from asset returns.
2. Perform eigenvalue decomposition to obtain eigenvalues and eigenvectors/
3. Compare each eigenvalue to the Marchenko-Pastur upper bound.
4. Treat eigenvalues within the noise range as unstable and replace them with a smoothed estimate.
5. Preserve eigenvalues that exceed the threshold, as they capture significant relationships.
6. Reconstruct a cleaned correlation matrix using the adjusted eigenvalues.

Once the denoised correlation matrix is obtained, the model rebuilds the covariance matrix using the estimated asset volatilities:

$$\Sigma_{clean} = D \times Corr_{clean} \times D$$

Where D is the diagonal matrix of asset volatilities.

This process reduces the impact of spurious correlations while preserving the dominate structure in the data. As a result, the optimization becomes more stable and less sensitive to small changes in input data.

The denoising procedure is based on Marchenko-Pastur Distribution, which separates signal from noise in empirical correlation matrices. By incorporating Marchenko-Pastur filtering, the model improves robustness and reliability of portfolio construction. This step is especially important in higher0dimensional settings, where noisy estimates can otherwise lead to extreme or poorly diversified allocations. While this model uses spectral denoising, alternative approaches such as shrinkage-based covariance estimation provide a complementary method for improving stability and represent a potential extension.

3.4 Portfolio Optimization Framework – The portfolio optimization model is based on Modern Portfolio Theory, originally developed by Harry Markowitz. The model maximizes a risk adjusted utility faction that trades off expected return and portfolio variance. The original version of the model uses a constrained mean-variance optimization framework to determine portfolio allocations across equities, cash, and Treasury bills. This baseline implementation follows the standard Markowitz formulation and serves as the foundation for the initial OOS (Out of sample) back testing results. The optimization problem is defined by the objective:

$$U(w) = \mu^T w - \frac{1}{2}\lambda w^T \Sigma w$$

Where w represents the portfolio weights, μ is the vector of expected returns estimated from historical data, and Σ is the covariance matrix derived from asset returns. The first term represents expected portfolio return, while the second term penalizes total portfolio variance.

In the baseline model, the risk-aversion parameter λ directly controls the tradeoff between return and risk. Lower values of λ lead to more aggressive portfolios that prioritize expected return, while higher values place greater emphasis on risk reduction and diversification. The portfolio is constructed across 3 components: equities, cash, and Treasury bills form a defensive allocation. Cash provides liquidity, while Treasury bills offer a low-risk yield alternative to idle capital.

The optimization is subject to several constraints. All portfolio weights must sum to 1, ensuring that all capital is allocated. The model enforces a long-only constant, preventing short positions. Each stock is also subject to a maximum allocation cap, which is either specified directly by the portfolio manager or applied uniformly across the portfolio.

The defensive sleeve is controlled through minimum and maximum allocation constraints of the combined weight of cash and T-Bills. This ensures that the portfolio maintains a required level of capital preservation while still allowing flexibility in how that defensive allocation is distributed.

The model also supports optional constraints such as a minimum expected return or maximum allowable volatility. These constraints allow the optimizer to target specific portfolio characteristics, though they can reduce feasibility if the required thresholds are too restrictive.

The optimization is soled numerically using constrained nonlinear methods. The model searches over feasible portfolio weights to identify the allocation that maximizes the objective function while satisfying all constraints.

This baseline model produces a set of optimal portfolio weights based on historical return estimates and covariance structure. However, because it relies heavily on raw statistical estimates, it is sensitive to estimation error in both expected returns and correlations.

This sensitivity is reflected in the original OOS back testing results. While the model frequently produced portfolios with positive realized returns, the expected-return forecasts were often overly optimistic. As a result, many portfolios were profitable in absolute terms but failed to meet the model's predicted return threshold, leading to relatively low formal prediction accuracy despite strong profitability rates.

3.5 Portfolio Construction & Implementation - The optimization process produces a vector of continuous portfolio weights, but these weights are not directly implementable in practice. Real portfolios must be expressed in discrete units, such as whole shares, and must account for capital constraints and residual cash balances. This section describes how the model translates theoretical allocations into an executable portfolio.

The first step is to convert optimized weights into dollar allocations. Given total portfolio capital, each asset's target dollar position is calculated by multiplying its weight by total capital. This produces a continuous target allocation across all assets, including equities, cash, and T-Bills.

For equities, these dollar allocations are then converted into discrete share counts. Since fractional shares are not assumed, each position is rounded down to the nearest whole share. This rounding process introduces small deviations for the original optimized weights, as the exact target allocation may not be perfectly achievable.

After rounding, the model computes the remaining unallocated capital. This residual capital arises from the difference between the continuous allocation and discrete share implementation. Rather than leaving this capital unused, the model reallocates it in a structured way.

First, the required defensive sleeve is satisfied. The model ensures that the combined allocation to cash and T-Bills meets the minimum constraint specified by the portfolio manager. Within this sleeve, Treasury bills are used to generate yield on defensive capital, while cash is retained for liquidity purposes.

Any remaining capital beyond the minimum defensive requirement is then distributed across equity positions. The model prioritizes assets where the gap between the target allocation and the implemented allocation is largest. This helps reduce tracking error between the optimized portfolio and the final implemented version.

The final output of this model is a fully specified portfolio including:

1. Exact share counts for each stock
2. Dollar allocation per asset
3. Total allocation to cash
4. Total allocation to T-Bills

This implementation step is important as it bridges the gap between theoretical optimization and real-world execution. While small deviations from the optimal weights are unavoidable due to discrete constraints, the model ensures that these deviations are minimized and handled systematically. By incorporating portfolio construction directly into the framework, the model produces results that are not only optimal in theory but also verified on real world information.

3.6 Regime Switching Monte Carlo Simulation (HMM) – The model evaluates forward portfolio performance using a regime-switching Monte Carlo simulation driven by a Hidden Markov Model (HMM). This framework allows return distributions to vary over time instead of assuming a single stationary process.

Regime Identification

The model constructs a market factor using the cross-sectional average of asset returns. This time series is used to fit a Gaussian hidden Markov Model, which assumes returns are generated by a finite set of latent regimes. Each regime is defined by:

1. Mean return
2. Variance
3. Transition Probabilities

The HMM estimates these parameters from historical data and infers the current regime probabilities.

Regime-Specific Return Estimation

After regime identification, the model partitions historical returns by regime. For each regime k, it computes μ^k, Σ^k where μ^k is the regime-specific expected return vector and Σ^k is the regime-specific covariance matrix. This allows the model to represent different market conditions, such as low-volatility and high-volatility environments, each with distinct return characteristics.

Monte Carlo Simulation Process

The simulation generates future portfolio paths using regime transitions stochastic return draws. For each simulation path:

1. Initialize the starting regime using current regime probabilities.
2. At each time step:
    a. Sample the next regime using the Markov transition matrix
    b. Draw asset returns from a multivariable normal distribution:
        i. $r_t \sim N(\mu^k, \Sigma^k)$ where k is the current regime.

3. Update portfolio value.
4. Repeat over the full simulation horizon.

This process is repeated across multiple simulation paths to generate a distribution of outcomes.

Portfolio Evolution

At each step:

1. Equity positions evolve based on simulated returns
2. Cash accrues its specified yield
3. T-Bills accrue their assigned yield

Total portfolio value is updated sequentially across the simulation horizon

Simulation Outputs

After all paths are generated, the model summarized terminal outcomes including expected terminal value, median terminal value, percentile ranges, expected returns over the horizon, and probability of loss. These metrics describe the distribution of potential outcomes rather thana single estimate.

Role in the Framework

The optimization step produces a single allocation based on historical estimates. The monte Carlo layer evaluates that allocation under varying market conditions. By incorporating regime switching, the model avoids assuming constant volatility and correlation. Instead, it reflects changing market environments, producing a more realistic assessment of portfolio risk and return.

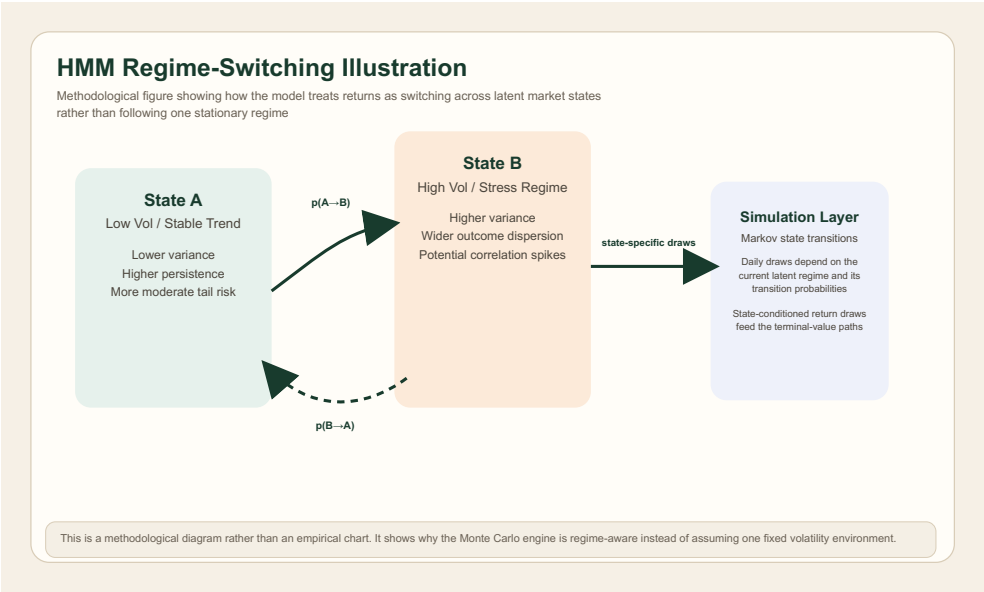

Figure 1

Hidden Markov Model regime identification illustrating transitions between market states used in simulation

The set underlying this framework includes both methodologies that are directly implemented in the model and those informing its development path. The current implementation revolves around the Modern Portfolio Theory, with covariance denoising based on the Marchenko-Pastur Distribution and regime dynamics modeled using a Hidden Markov Model. Expected return estimation incorporates both Bayes-Stein Estimation and Black-Litterman Model approaches. Additional references, including covariance shrinkage methods, factor-based return models, volatility managed portfolios, and Bayesian allocation, are included to support future extensions of the model.

4. Monte Carlo Simulations – The Monte Carlo simulation produces a distribution of possible portfolio outcomes over the selected investment horizon. This distribution is generated using regime-switching return dynamics and reflects both return uncertainty and changes in market conditions.

For each optimized portfolio, the model runs a specified number of simulation paths (5k+). Each path represents a possible evolution of portfolio value based on stochastic returns and regime transitions. The result is a full distribution of terminal portfolio values rather than a single estimate.

The model reports key summary statistics from this distribution. These include the expected terminal value, median outcome, and multiple percentile levels that describe the range of possible results. Lower percentiles represent downside scenarios, while upper percentiles capture favorable outcomes.

One of the primary risk measures is the probability of loss, defined as the proportion of simulation paths where the final portfolio value is below the initial capital. This metric provides a direct estimate of downside risk over the selected horizon.

The simulation results often show a wide dispersion between lower and upper outcomes, reflecting the combined impact of return volatility, correlation structure, and regime transitions. In higher-volatility regimes, the distribution widens, increasing both upside potential and downside risk.

A key observation from the simulation output is that expected return alone is not sufficient to evaluate portfolio quality. Portfolios with similar expected returns can have significantly different downside profiles. The Monte Carlo framework captures this difference by explicitly modeling the distribution of outcomes.

In the baseline model, simulation results frequently indicated positive expected returns and a high likelihood of profitability. However, the dispersion of outcomes and probability of

underperformance relative to expected return highlighted limitations in return estimation. Many portfolios produced positive realized return but failed to meet the model's forecasted return level.

This reinforces the importance of evaluating portfolios using distribution-based metrics rather than relying on expected return and variance. The Monte Carlo simulation provides a more complete view of risk by incorporating both uncertainty and regime-dependent behavior.

Overall, this component of the model transforms static portfolio estimates into a forward-looking distribution, allowing for a more realistic assessment of potential performance.

5. System Architecture - The system is organized as a modular pipeline that processes user inputs, constructs the portfolio, and evaluates forward performance. Each component is separated by function to ensure clarity, flexibility, and ease of modification.

Data Layer - The system begins by retrieving historical adjusted-close price data for all user-specified tickers. This data is used to computer return series, volatility, covariance, and correlation matrices. Data preprocessing includes aligning time series, handling missing values, and converting prices into daily log returns.

Estimation layer - From the processed return data, the model estimates expected returns and risk metrics. This includes:

1. Annualized expected return vector
2. Covariance Matrix
3. Correlation Matrix

The correlation matrix is then denoised using the Marchenko-Pastur procedure to produce a cleaned covariance structure. These estimates form the core inputs for the optimization stage

Optimization Layer - The optimization layer solves the constrained mean-variance problem using the estimated inputs. It determines portfolio weights across equities, cash, and T-Bills while enforcing all constraints, including:

Long-Only constraint

1. Maximum allocation per asset
2. Defensive sleeve constraints
3. Optional return and volatility targets

The output of this layer is a vector of optimal portfolio weights.

Implementation Layer - The implantation layer converts continuous weights into an executable portfolio. Target weights are translated into dollar allocations and then into discrete share counts. Residual capital is allocated across cash and T-Bills, and any remaining funds are redistributed to minimize deviation from target allocations.

Simulation Layer - The simulation layer evaluates forward performance using the regime-switching Monte Carlo framework. A Hidden Markov Model is fitted to market data, and regime-dependent return distributions are used to simulate portfolio evolution over time.

This layer produces a distribution of possible outcomes and calculates risk metrics such as probability of loss and percentile ranges.

Output Layer - The final output combines optimization results and simulation metrics into a structured report. The system provides:

1. portfolio allocations and share counts
2. expected return and volatility
3. Monte Carlo distribution statistics
4. probability of loss
5. portfolio risk score and classification

This architecture separates data processing, estimation, optimization, and simulation into distinct components. This design improves transparency, allows for targeted improvements, and supports future extensions without restructuring the entire system.

6. Example Results / Tested Results

6.1 Back testing Design and Evaluation Methodology - The results are generated using a point-in-time, OOS back testing framework.

1. A formation date is selected.
2. A 1-year lookback window is used to estimate expected returns and covariance using only historical data available at that date.
3. A portfolio is constructed based on these estimates and model constraints.
4. The portfolio is held over a one-year forward period.

This process is repeated across multiple randomly generated portfolio combinations. Each portfolio contains a fixed number of stocks entered from the defined universe and is subject to allocation constraints. For each simulation, the model records expected return, realized return, Monte Carlo expected return, and terminal portfolio value. Prediction accuracy is defined as the proportion of portfolios where realized return is greater than or equal to expected return. This metric evaluates calibration of the return estimator rather than profitability. All reported results are based on 994 completed simulations with 6 failed runs.

6.2 Baseline Model Results - The baseline model uses the historical mean return estimator. Across 994 simulations, the model produced a prediction accuracy of 23.44%, a mean expected return of 19.34%, and a mean realized return of 9.78%. The results show consistent overestimation of expected returns. Most portfolios generated positive realized returns but failed to meet their predicted return thresholds, indicating that the optimization process is effective while return estimation is not trustworthy.

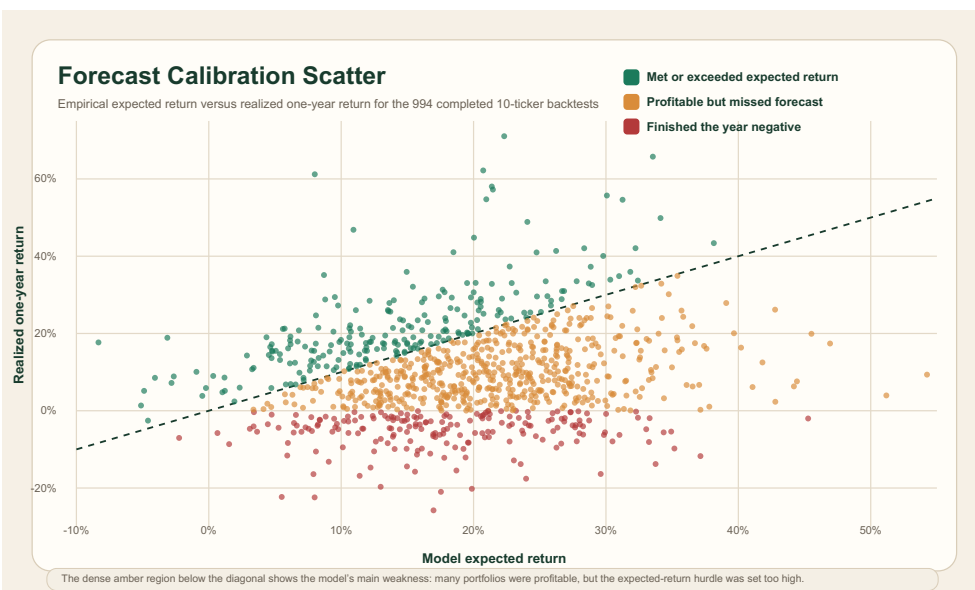

Figure 2

Figure X. Forecast calibration scatter comparing expected and realized returns across portfolio simulations. The deviation from the diagonal reflects systematic overestimation.

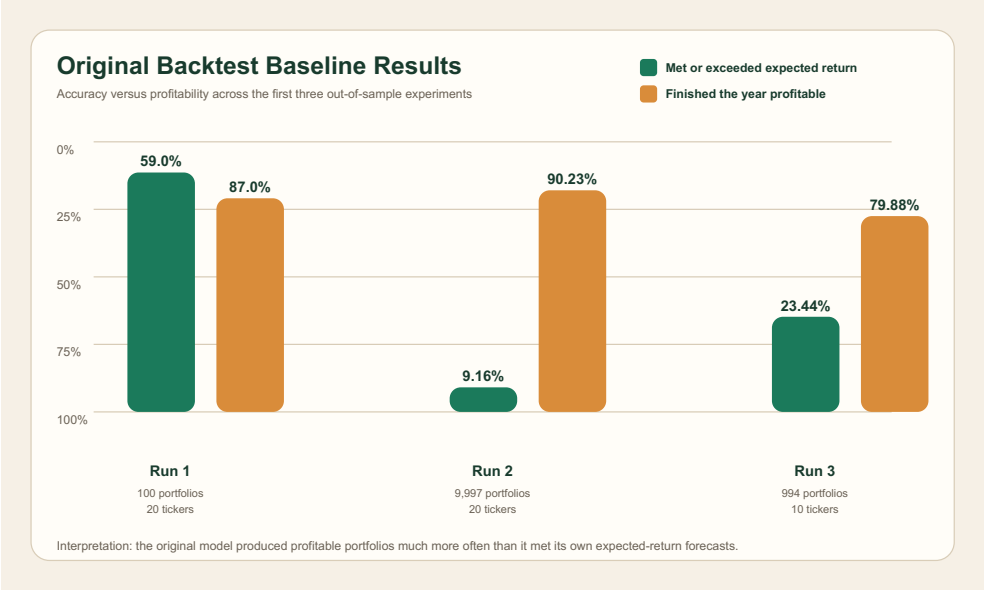

Figure 3

Figure X. Baseline back test results summarizing portfolio performance across simulations.

6.3 Updated Model Results (Return Estimator Comparison) - The historical mean estimator completed 994 simulations with 6 failures and produced an accuracy of 23.44%, a mean expected return of 19.34%, and a mean realized return of 9.78%. The Bayes-Stein estimator completed 994 simulations with 6 failures and produced an accuracy of 23.24%, a mean expected return of 19.30%, and a mean realized return of 9.79%, indicating minimal improvement from shrinkage. The market-factor estimator completed 994 simulations with 6 failures and produced an accuracy of 76.76%, a mean expected return of 1.04%, and a mean realized return of 9.12%. The higher accuracy is driven by a significantly lower expected-return threshold. The Black-Litterman estimator is based on the framework developed by Fischer Black and Robert Litterman, which combines prior market equilibrium returns with adjusted expectations. The Black-Litterman estimator completed 994 simulations with 6 failures and produced an accuracy of 46.48%, a mean expected return of 10.14%, a mean Monte Carlo expected return of 61.24%, a mean realized return of 9.77%, and a median realized return of 8.83%. This estimator reduces return bias and improves alignment between expected and realized performance.

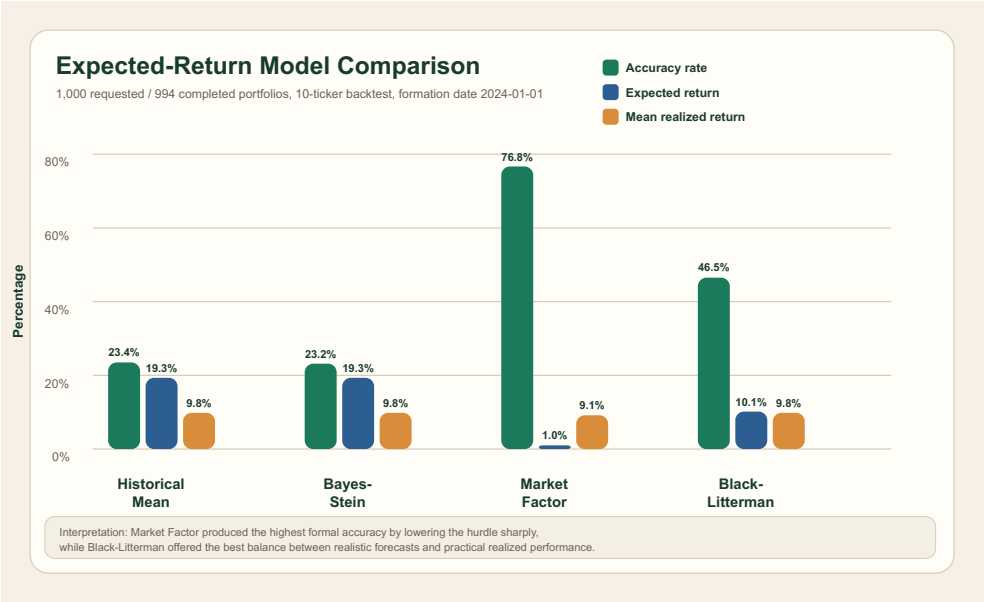

Figure 4

Comparison of prediction accuracy and expected-return calibration across return estimators.

6.4 Comparison of Results - The results show a direct relationship between expected-return magnitude and prediction accuracy.

1. Estimators with higher expected returns produce lower accuracy due to overestimation.
2. Estimators with lower expected returns produce higher accuracy due to easier thresholds.
3. Estimators with moderate expected returns improve calibration while maintaining economic relevance.

All estimators produce similar realized returns, indicating that portfolio construction is stable and that performance differences are driven by return estimation.

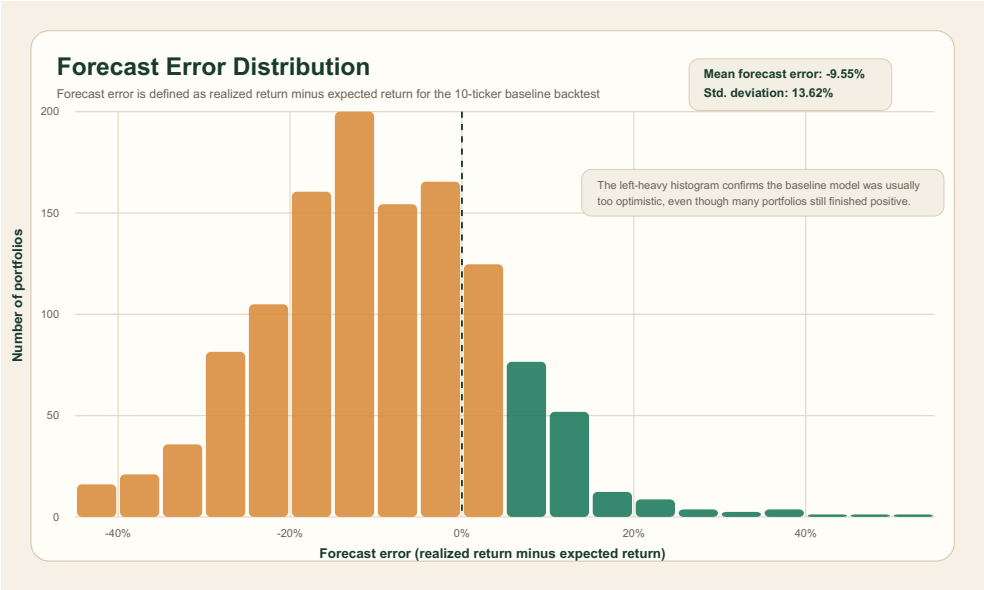

Figure 5

Distribution of forecast errors (realized minus expected returns), showing systematic bias and dispersion.

6.5 Interpretation of Results - The results indicate that expected-return estimation is the primary limitation of the model. The historical mean estimator produces inflated forecasts, reducing accuracy despite positive realized outcomes. The market-factor estimator improves accuracy by lowering the expected-return threshold but reduces usefulness for return targeting. The Black-Litterman estimator improves calibration by incorporating structured priors, resulting in closer alignment between expected and realized returns. Improvements in return estimation have a greater impact on performance than changes to optimization or simulation.

7. Portfolio Interpretation

The model produces allocation outputs and distribution-based risk metrics. Risk is evaluated using:

1. Portfolio volatility
2. Probability of loss derived from simulation
3. Distribution dispersion measured through percentile ranges.

These metrics are combined to generate a risk score and classification. Monte Carlo results provide a distribution of terminal outcomes where lower percentiles represent downside risk and upper percentiles represent potential upside. Portfolios with similar expected returns can have different risk profiles due to differences in volatility and correlation. This is reflected in variation in probability of loss and distribution spread. Allocations to cash and Treasury bills reduce downside risk and stabilize outcomes, particularly under higher-volatility regimes.

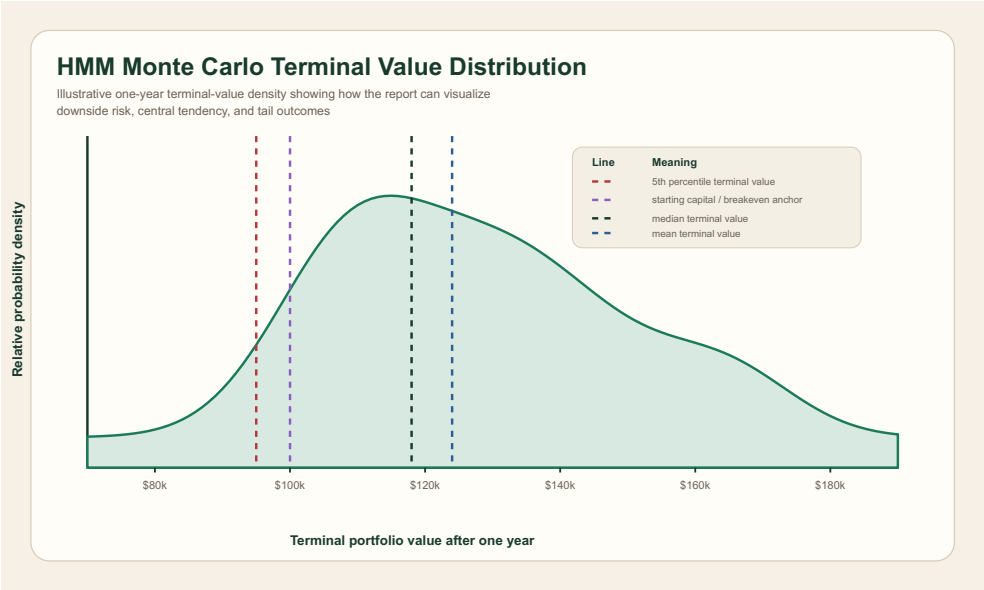

Figure 6

Monte Carlo distribution of simulated portfolio outcomes, illustrating dispersion and downside risk.

8. Limitations

The model relies on historical data for parameter estimation, which may not reflect future conditions. Expected-return estimation is the primary source of error, and small deviations can significantly affect optimization results. Correlation denoising reduces noise but does not eliminate estimation risk, especially during market stress when relationships between assets may change. The Hidden Markov Model assumes a finite number of regimes and may not capture all market dynamics. The model does not account for transaction costs, taxes, slippage, or liquidity constraints. Converting continuous weights into discrete share allocations introduces small

deviations from optimal weights. The framework is limited to long-only portfolios and does not include leverage or hedging strategies.

The model also does not incorporate shrinkage-based covariance estimation, multi-factor expected-return models, or volatility-managed allocation overlays. While constraints and denoising improve stability, estimation error remained a central limitation. These limitations are consistent with prior findings that optimized portfolios can underperform simpler allocation strategies when inputs are noisy, reinforcing the importance of robustness and benchmarking in portfolio construction.

9. Conclusion and Final Thoughts

This project developed a portfolio optimization model that integrates return estimation, constrained optimization, and regime-based simulation. Back testing results show that the model consistently generates portfolios with positive realized returns, but the baseline implementation suffers from poor calibration due to overestimation of expected returns. Testing alternative estimators shows that improving return estimation significantly enhances performance. The Black-Litterman estimator provides the most balanced improvement by increasing prediction accuracy while maintaining realistic expected returns. The development process follows an iterative structure where model improvements are driven by empirical results. Future work should focus on improving return priors, incorporating factor-based models, and extending the framework to include market-neutral strategies and transaction cost modeling. The final system provides a structured and extensible approach to portfolio construction. This model builds on established financial theory while extending it through improved estimated techniques and simulation-based evaluation.

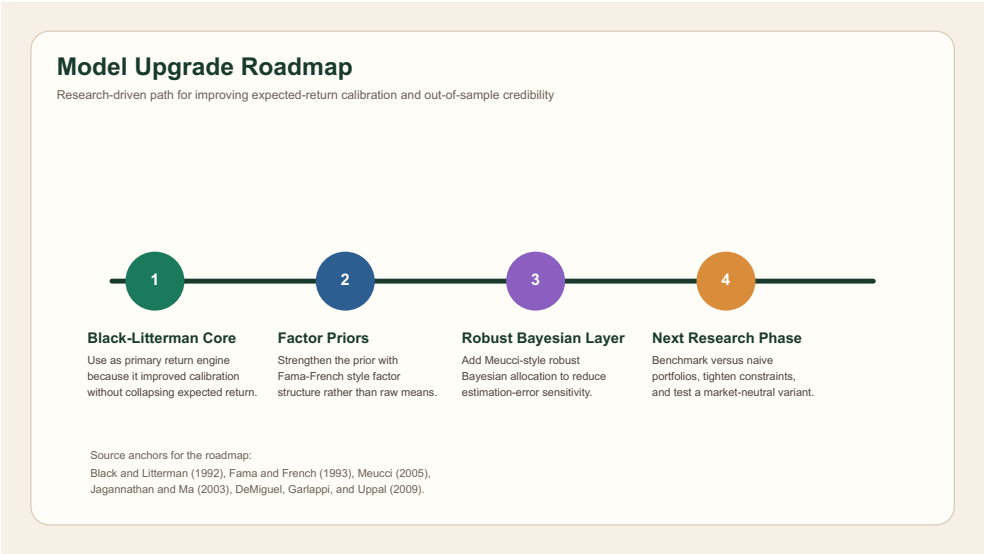

Figure 7

Model development roadmap showing profession from baseline optimization to improved estimation and simulation models.

The model reflects an iterative development process in which classical portfolio theory is extended through improved estimation techniques and simulation-based evaluation. While the current model integrates several established methodologies, additional enchantments -such as covariance shrinkage, factor-based return modeling, volatility managed exposure, and robust Bayesian allocation- provide a clear path to a more stable and production-ready system.

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